

Intro to NT

Monday, 2 June, 2025 8:20 AM

Number Theory

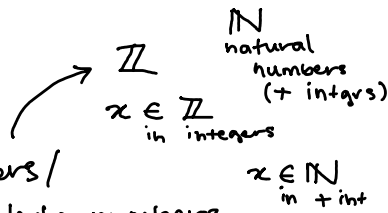
$$\begin{cases} 5.2 \\ 6 \nmid 7 \end{cases}$$

integers/
whole numbers

number bulat

$$5.2 \quad 6 \nmid 7$$

$$\textcircled{7} \quad \nmid \quad 41.9$$



1: DIVISIBILITY

$$a \mid b \Leftrightarrow b \text{ is div. by } a = a \text{ div. } b$$

$$k = \frac{b}{a} \in \mathbb{Z} \quad 2 \mid 6 \quad \frac{6}{2} = 3 \in \mathbb{Z}$$

$$2 \nmid 5 \quad \frac{5}{2} = 2.5 \notin \mathbb{Z}$$

$$\begin{cases} x \mid 0 \quad T \quad x \neq 0 \quad \frac{0}{x} = 0 \in \mathbb{Z} \\ x \in \mathbb{Z} \quad x = 0 \quad F \quad \frac{0}{0} = ? \quad X \\ 0 \mid x \quad F \end{cases}$$

$$x \mid y, y \mid z \rightarrow \frac{y}{x}, \frac{z}{y} \in \mathbb{Z}$$

$$\begin{aligned} y &= ax, a \in \mathbb{Z} \\ z &= by, b \in \mathbb{Z} \\ &= b(ax) \\ &= (ba)x \quad ba \in \mathbb{Z} \\ &\Rightarrow x \mid z \quad \text{proven} \end{aligned}$$

Very important:

$$a \mid \text{smth} \rightarrow \text{can subtract multiples of } a$$

$$\begin{aligned} \rightarrow a \mid a+b &\rightarrow a \mid b \\ \hookrightarrow a+b &= ac \\ b &= ac-a = a(c-1) \\ &\in \mathbb{Z} \end{aligned}$$

$$\begin{aligned} a \mid b &\Leftrightarrow \frac{b}{a} = k, k \in \mathbb{Z} \\ &\uparrow \\ &|a| \leq |b| \quad b \neq 0 \\ &b=0 \\ &\rightarrow |b| = |k||a| \geq |a| \\ &k \neq 0, |k| \geq 1 \end{aligned}$$

remainder bounding

$$\begin{aligned} x \mid y &\rightarrow \frac{y}{x} \in \mathbb{Z} \\ \text{prove } x \mid ay &\quad a \in \mathbb{Z} \\ y &= bx, b \in \mathbb{Z} \\ ay &= (ab)x \rightarrow x \mid ay \quad \text{proven} \end{aligned}$$

$$\begin{aligned} x \mid y, x \mid z &\rightarrow \frac{y}{x}, \frac{z}{x} \in \mathbb{Z} \\ \text{prove } x \mid ay + bz &\quad \frac{ay+bz}{x} \\ y &= cx, z = dx \\ ay + bz &= a(cx) + b(dx) \\ &= x(ac+bd) \\ &\in \mathbb{Z} \quad \text{proven} \end{aligned}$$

$$a > b > 0$$

prove $a \nmid a+b$ is impossible

$$a \mid 5a+b \rightarrow a \mid 4a+b \rightarrow a \mid 3a+b \dots \rightarrow a \mid b$$

$$a \mid b \rightarrow \cancel{b \mid a} \rightarrow \text{impossible}$$

$n > 1$

prove

$$n \mid 20n^2 + 25n + 1$$

is impossible

$$\rightarrow n \mid 1 \text{ impossible}$$

Find all $n \in \mathbb{N}$ s.t.

$$n \mid 20n + 25$$

$$\rightarrow n \mid 25$$

$$\rightarrow 1, 5, 25$$

2: PRIMES

p is prime

$\leftrightarrow$ divisors are only 1 and p

$\times$

$\neq$ inf. many primes

euclid's proof

• assume finite primes $\rightarrow k$

$$p_1, p_2, \dots, p_k$$

$$p_1 p_2 \dots p_k + 1 = N$$

$\rightarrow$ prime

$$\rightarrow p_i$$

$$p_i \mid p_1 p_2 \dots p_k + 1$$

$$\rightarrow p_i \mid 1 \rightarrow p_i > 1$$

Find all $a \in \mathbb{N}$

s.t. $a^2 - 25$ is prime

$$a = 6$$

$$a^2 - 25 = p$$

$$a = 6 \quad p = 11$$

$$\rightarrow a^2 - 5^2 = (a-5)(a+5) = p$$

$$a^2 - 49 \text{ no sol.}$$

$$(a-7)(a+7) = p$$

$$a = 8 \quad p = 15 \quad \times$$

$$p, q \text{ odd}$$

$$p+q = 85$$

find all (p, q)

$$(2, 83)$$

$$(83, 2)$$

2 only even prime

odd + odd = even

3: PRIME FACTORISATION

Fundamental Theorem of Arithmetic (FTA)

every $n > 1 \in \mathbb{Z}$

has a unique

prime factorisation

$$15 = 3 \times 5$$

$$25 = 5^2$$

$$24 = 2^3 \times 3$$

$$300 = 2^2 \times 3 \times 5^2$$

$$120 = 2^3 \times 3 \times 5 = 5 \times 2^3 \times 3$$

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$$

$p_i = \text{primes}$

$a_i = \text{non-neg int'gs}$

$$\begin{array}{c} 120 \\ \swarrow \searrow \\ 2 \quad 60 \\ \quad \swarrow \searrow \\ \quad 2 \quad 30 \\ \quad \quad \swarrow \searrow \\ \quad \quad 2 \quad 15 \\ \quad \quad \quad \swarrow \searrow \\ \quad \quad \quad 2 \quad 5 \end{array}$$

$$N = 2^4 \times 3^5 \times 5^3$$

$$x \mid N \quad x = 2^a \times 3^b \times 5^c$$

$$0 \leq a \leq 4$$

$$0 \leq c \leq 3$$

$$N = 2^5$$

$$2^0, 2^1, \dots, 2^4, 2^5$$

$$2^6 \times$$

$$N = p_1^{a_1} \times p_2^{a_2} \times \dots \times p_k^{a_k}$$

how many positive divisors of N ?

★ why?

$$x|N \quad x = p_1 \frac{b_1}{I} \times p_2 \frac{b_2}{I} \times \dots \times p_k \frac{b_k}{I} \quad \begin{matrix} \downarrow \\ a_{k+1} \end{matrix}$$

$$(a_1 + 1)(a_2 + 1) \cdots (a_k + 1)$$

$$x|N \quad x = 2^a \times 3^b \times 5^c$$

$$0 \leq a \leq 4 \quad 0 \leq c \leq 3$$

$$0 \leq b \leq 5$$

$$\frac{2^4 \times 3^5 \times 5^3}{2^2}$$

perfect cube

perfect squares

$5^2 \quad 35^2$

$$2^4 \times 3^2 \times 7^2 \times 11^2$$
$$= (2^2 \times 3 \times 7 \times 11)^2$$

$$\begin{aligned} & \dots, 2^4, \\ & \quad 2^5 \\ & 2^4 \times \\ & \frac{2^5}{2^4} = \frac{1}{2} \times \end{aligned}$$

$$\begin{array}{l} 3^6 \times 5^3 \\ 3^5 \times 5^3 \end{array}$$

→ FSTB
greatest common divisor (GCD)

$$\gcd(16, 20) = 4$$

$$A = \begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 3 & 5 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 5 & 2 \\ 5 & 5 \end{bmatrix}$$

$$\gcd(A, B) = 2^4 \times 3^2 \times 5$$

$$d = p_1^{a_1} \times p_2^{a_2} \times \dots \times p_k^{a_k}$$

$$\beta = p_1^{b_1} \times p_2^{b_2} \times \dots \times p_k^{b_k}$$

$$\gcd(\alpha, \beta) = p_1^{\min(a_1, b_1)} \times p_2^{\min(a_2, b_2)} \times \dots \times p_k^{\min(a_k, b_k)}$$

least common multiple (LCM)

$$lcm(3, 5) = 15$$

$$lcm(2, 4) = 4$$

$$A = 2^4 \times 3^5 \times 5^2$$

$$B = 2^5 \times 3^3 \times 5$$

$$\rightarrow \text{lcm}(A, B) = 2^5 \times 3 \times 5^2$$

$$\alpha = p_1^{a_1} \times p_2^{a_2} \times \dots \times p_k^{a_k}$$

$$\beta = p_1^{b_1} \times p_2^{b_2} \times \dots \times p_k^{b_k}$$

$$\text{lcm}(a, \beta) = p_1^{\max(a_1, b_1)} \times p_2^{\max(a_2, b_2)} \times \dots \times p_k^{\max(a_k, b_k)}$$

Find the no. of positive factors of 2025 which are perfect

$$2025 = 3^4 \times 5^2 \quad \text{cubes}$$

$x \mid 2025$
 $3 \mid x$

$x = 3^a \times 5^b$
 $1 \leq a \leq 4, 0 \leq b \leq 2$
 $\square \quad 4 \times 3 = \boxed{12}$
 $a = 0, 2, 4 \rightarrow 3^x$
 $\square \quad b = 0, 2 \rightarrow 2^x = \boxed{6}$
 $a = 0, 3 \rightarrow 2$
 $b = 0 \rightarrow 1^x = \boxed{2}$
 $3^0 \times 5^0 \quad 3^3 \times 5^0$
 $1 \quad 27$

Find smallest $m \in \mathbb{N}$

s.t - $2m = \text{perfect square}$

$3m = \text{perfect cube}$

$$\begin{aligned} m &= 2^a \times 3^b & a &= 3 \\ 2m &= 2^{a+1} \times 3^b & b &= 2 \\ 3m &= 2^a \times 3^{b+1} \\ m &= 2^3 \times 3^2 = 72 \end{aligned}$$

Prove that $n \in \mathbb{N}$ has an

Prove that $n \in \mathbb{N}$ has an odd number of positive factors

if and only if n is a perfect square.

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$$

$$(a_1+1)(a_2+1) \dots (a_k+1) = \text{odd}$$

$$a_i+1 = \text{odd} \quad a_i = \text{even}$$

$$a_i = \text{even} \quad \rightarrow n = x^2$$

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$$

$$a_i = \text{even}$$

$$(a_1+1)(a_2+1) \dots (a_k+1) = \text{odd}$$

proven

Prove $ab = \gcd(a, b) \times \text{lcm}(a, b)$

$$\{c_i, d_i\}$$

$$a = p_1^{c_1} p_2^{c_2} \dots p_k^{c_k}$$

$$b = p_1^{d_1} p_2^{d_2} \dots p_k^{d_k}$$

$$ab = p_1^{c_1+d_1} \dots$$

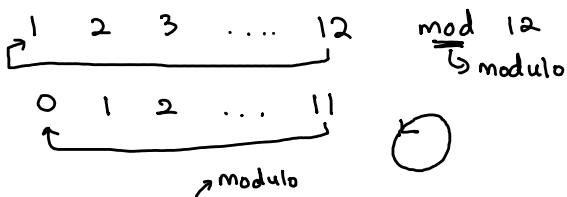
$$\gcd(a, b) = p_1^{\min(c_1, d_1)} p_2^{\min(c_2, d_2)} \dots p_k^{\min(c_k, d_k)}$$

$$\text{lcm}(a, b) = p_1^{\max(c_1, d_1)} p_2^{\max(c_2, d_2)} \dots p_k^{\max(c_k, d_k)}$$

$$\frac{c_i+d_i}{\min(c_i, d_i)} = \{ \min(c_i, d_i), \max(c_i, d_i) \}$$

$$p_1^{\min(c_1, d_1)} + \max(c_1, d_1)$$

4: MODULAR ARITHMETIC



$$a \equiv b \pmod{n} \Leftrightarrow n \mid a-b$$

congruent

$$7 \equiv 4 \pmod{3} \equiv 1$$

residue

$$3 \equiv 5 \pmod{2}$$

$$2 \mid 5-3=2 \checkmark$$

$$5 \pmod{3} \equiv 2$$

$$7 \pmod{5} \equiv 2$$

$$4 \pmod{2} \equiv 0$$

$$35 \pmod{3} \equiv 2$$

$$35 \equiv 5 \times 7 \pmod{3}$$

$$\equiv 2 \times 1 \pmod{3}$$

$$\equiv 2 \pmod{3}$$

① addition $q \pmod{3} \equiv 0$

$$4+5 \pmod{3}$$

$$\equiv 1+2 \pmod{3}$$

$$\equiv 3 \pmod{3}$$

$$\equiv 0 \pmod{3}$$

③ multiply

$$4(5) \pmod{3}$$

$$\equiv 1(2) \pmod{3}$$

$$\equiv 2 \pmod{3}$$

② subtract

useless

$$2025 \text{ when div by } 5$$

find remainder

when 42025 is divided by 3

$$20 \equiv 2 \pmod{3}$$

$$2 \equiv 2 \pmod{3}$$

$$2 \equiv -1 \pmod{3}$$

perfect squares mod 3

1, 4, 9, 16, 25, 36

mod 4

1, 0, 1, 0, 1, 0, ...

that remainder

when 4^{2025} is divided by 3

$$4 \equiv 1 \pmod{3}$$

$$\begin{aligned} 4^{2025} \pmod{3} &\equiv 1 \\ &\equiv \underbrace{4 \times 4 \times \dots \times 4}_{2025} \pmod{3} \\ &\equiv \underbrace{1 \times 1 \times \dots \times 1}_{2025} \pmod{3} \\ &\equiv 1 \end{aligned}$$

$$\begin{aligned} 4^{2025} &\equiv 2^{4050} \pmod{3} \\ &\equiv (-1)^{4050} \pmod{3} \\ &\equiv 1 \pmod{3} \end{aligned}$$

$$2 \equiv -1 \pmod{3}$$

mod 3

1, 4, 9, 16, 25, 36

1, 1, 0, 1, 1, 0

$$\begin{aligned} 5^2 \pmod{3} &\equiv 5 \times 5 \pmod{3} \\ &\equiv 2 \times 2 \pmod{3} \\ &\equiv 1 \pmod{3} \end{aligned}$$

1, 0, 1, 0, 1, 0, ...

~~XX~~

$$\cancel{2} \pmod{4}$$

remainder when

7^{2025} is divided by 8

$$7^{2025} \pmod{8}$$

$$\begin{aligned} &\equiv 7(7^{2024}) \pmod{8} \\ &\equiv 7(49^{1012}) \pmod{8} \\ &\equiv 7(1^{1012}) \pmod{8} \\ &\equiv 7 \pmod{8} \end{aligned}$$

$$7^{2025} \pmod{8}$$

$$\begin{aligned} &\equiv (-1)^{2025} \\ &\equiv -1 \equiv 7 \end{aligned}$$

$$7^{2025} \pmod{5}$$

$$\equiv 7(7^{2024}) \pmod{5}$$

$$\begin{aligned} &\equiv 7(49^{1012}) \\ &\equiv 7(-1)^{1012} \\ &\equiv 7 \equiv 2 \end{aligned}$$

$$7^{2025} \pmod{9} = 7^3 = 343$$

$$\begin{aligned} &\equiv (7^3)^{675} \pmod{9} \\ &\equiv 1^{675} \equiv 1 \end{aligned}$$

divisibility rules

div by 9 $\leftrightarrow$ sum of digits

$$\begin{aligned} \overline{abc} &\equiv 100a + 10b + c \pmod{9} \\ &\equiv a + b + c \pmod{9} \end{aligned}$$

powerful for

★ deriving contradictions

$[2 + 5^n \text{ is NEVER a perfect square}]$

0, 1, 4, 9, 16, 25

$$\begin{aligned} 2 + 5^n \pmod{10} &\rightarrow \leftarrow \\ &\equiv 2 + 5 \equiv 7 \pmod{10} \end{aligned}$$

$$2 + 5^n \pmod{5} \equiv 2$$

2 0 1 2 3 4

2² 0 1 4 4 1 m5

$$2 + 5^n \pmod{4} \equiv 2 + 1^n \equiv 2 + 1$$

$$\equiv 3 \pmod{4}$$

$$\begin{aligned} [x^2 + y^2 = 3^{2025}] &\equiv 3(3^{2024}) \\ \text{no integer soln.} &= 3(9^{1012}) \\ &\equiv 3 \pmod{8} \end{aligned}$$

	0	1	2	3	4	5	6	7
$a^2 \pmod{8}$	0	1	4	1	0	1	4	1

no.

m4

$$\begin{aligned} n! &= 1 \times 2 \times \dots \times n \\ k! &= 1 \times 2 \times \dots \times k \end{aligned}$$

$$\begin{aligned} a^2 &= b! + 2023 \\ \text{no integer soln.} &\rightarrow 3 \end{aligned}$$

0/1

if $b \geq 4$,

then $b! \equiv 0 \pmod{4}$

$$b = \cancel{0}, \cancel{1}, \cancel{2}, \cancel{3}$$

$$45^2 = 2! + 2023$$

$$5 \pmod{3} \equiv 2$$

Prove $a^3 - a$

is a mult. of 6

$$a^3 - a = (a-1)(a)(a+1)$$

↓

a	$a^3 - a$
0	0
1	$1^3 - 1 \equiv 0 \pmod{6}$
2	$2^3 - 2 \equiv 6 \equiv 0 \pmod{6}$
3	$3^3 - 3 \equiv 24 \equiv 0 \pmod{6}$
4	

$$\begin{array}{l|l} 3 & 2-2=6 \equiv 0 \\ 4 & 3^3-3 \equiv 24 \equiv 0 \pmod{6} \\ 5 & \end{array}$$

$$5 \pmod{3} \equiv 2$$

$$2025 \pmod{4} \equiv 1$$

Find all (m, n) s.t.

$$1! + 2! + \dots + m! = n^2$$

$$\rightarrow 0/1/4 \pmod{5}$$

$$m \geq 5$$

$$1! + 2! + 3! + 4! \\ \equiv 1 + 2 + 6 + 24 \pmod{5} \\ \equiv 3 \rightarrow \leftarrow$$

$$m = 1, 2, 3, 4$$

$$m=1, n=1$$

$$m=2, n=2$$

$$m=3, n=3$$

$$m=4, n=4$$

JBMO 2014 P1

Find all triples of primes (p, q, r) satisfying

$$3p^4 - 5q^4 - 4r^2 = 26$$

$$3(5)^4 - 5(3)^4 - 4r^2 = 26$$

$$r^2 = 361$$

$$r = 19$$

$$(p, q, r) = (5, 3, 19)$$

$$3p^4 - 4r^2 \equiv 1 \pmod{5}$$

$$p^4 \equiv 0 \pmod{5} \rightarrow p \equiv 0 \pmod{5} \rightarrow \underline{p=5}$$

$$3p^4 + 5q^2 \equiv 1 \pmod{5}$$

$$r^2 \equiv 1 \pmod{5}$$

$$3(5)^4 - 5q^4 - 4r^2 = 26$$

$$-5q^4 - 4r^2 \equiv 2 \pmod{3}$$

$$q^4 + 2r^2 \equiv 2 \pmod{3}$$

$$q^4 \equiv 0 \pmod{3} \rightarrow \underline{q=3}$$

$$r^2 \equiv 1 \pmod{3}$$

$$5^2 \equiv 1 \pmod{3}$$